MOBILES AND SMART WATCHES ARE BANNED

RAMAIAH INSTITUTE OF TECHNOLOGY, BANGALORE-560054

DEPARTMENT OF CHEMISTRY

Code: CYC22

Sub: Engineering Chemistry

Course: II Sem B. E (all branches)

CIE Test: 1 **Term:** 01-03-2024 to 22-06-2024 **Max. Marks:** 30 **Time:** 60 Min

Credits: 3:0:0

Instructions: Answer any **two full** questions. Each question carries 15 marks

Q.No		Marks	CO's
1.	a. When iron is exposed to moist environment, it undergoes corrosion leading to the formation of rust. Explain the electrochemical theory of mechanism of corrosion by taking iron as an example.	5	CO2
	b. Justify: i) Copper gets deposited on iron nail immersed in $CuSO_4$ solution ii) Rate of corrosion is high in wet atmosphere than in dry atmosphere	5	CO2
	c. Define concentration cell. Calculate the cell potential of the following concentration cell: $Cu/Cu^{+2}(0.15\ M)//Cu^{+2}(0.23\ M)/Cu$ Comment on spontaneity of the cell.	5	CO1
2.	a. Explain: i) Capacity ii) Cycle life	5	CO1
	b. Derive the Nernst equation for the following electrode reaction: $Ag^+ + e^- \rightleftharpoons Ag$	5	CO1
	c. Justify: i) As Hydrogen overvoltage increases, rate of corrosion decreases ii) Larger cathodic area results in intense corrosion	5	CO2
3.	a. Justify the following: i) A mild steel tap in copper drum undergoes faster corrosion ii) A part of nail inside the wood undergo corrosion easily	5	CO2
	b. Write the construction and working of Lithium-ion battery	5	CO1
	c. Calculate the potential of Ag-Cu cell at 25°C, if the concentration of Ag^+ and Cu^{2+} are 0.01M and 0.08 M. Standard reduction potentials of Cu and Ag electrodes are +0.34 and +0.80 V. Calculate the change in free energy $\triangle G$ for the reduction of 1 mole of Ag^+ ions. Given that 1 Faraday =96.5 $KJV^{-1}\,mol^{-1}$	5	CO1